Terraform Cheat Sheet

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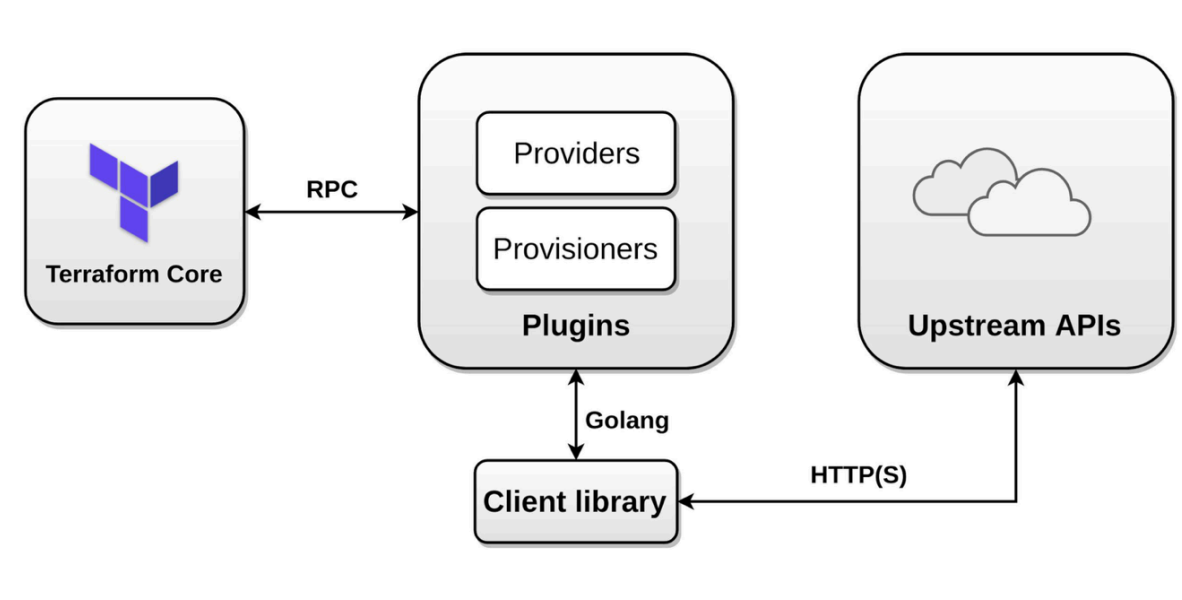
- An open source provisioning declarative tool that based on Infrastructure as a Code paradigm
- designed on immutable infrastructure principles
- Written in Golang and uses own syntax – HCL (Hashicorp Configuration Language), but also supports JSON
- Helps to evolve the infrastructure, safely and predictably
- Applies Graph Theory to IaaC and provides Automation, Versioning and Reusability
- Terraform is a multipurpose composition tool:
  ○ Composes multiple tiers (SaaS/PaaS/IaaS)
  ○ A plugin-based architecture model
- Terraform is not a cloud agnostic tool. It embraces all major Cloud Providers and provides common language to orchestrate the infrastructure resources
- Terraform is not a configuration management tool and other tools like chef, ansible exists in the market.

Terraform Architecture



Terraform Providers (Plugins)

- provide abstraction above the upstream API and is responsible for understanding API interactions and exposing resources.
- Invoke only upstream APIs for the basic CRUD operations
- Providers are unaware of anything related to configuration loading, graph theory, etc.
- supports multiple provider instances using `alias` *for e.g. multiple aws provides with different region*
- can be integrated with any API using providers framework
- Most providers configure a specific infrastructure platform (either cloud or self-hosted).
- can also offer local utilities for tasks like generating random numbers for unique resource names.

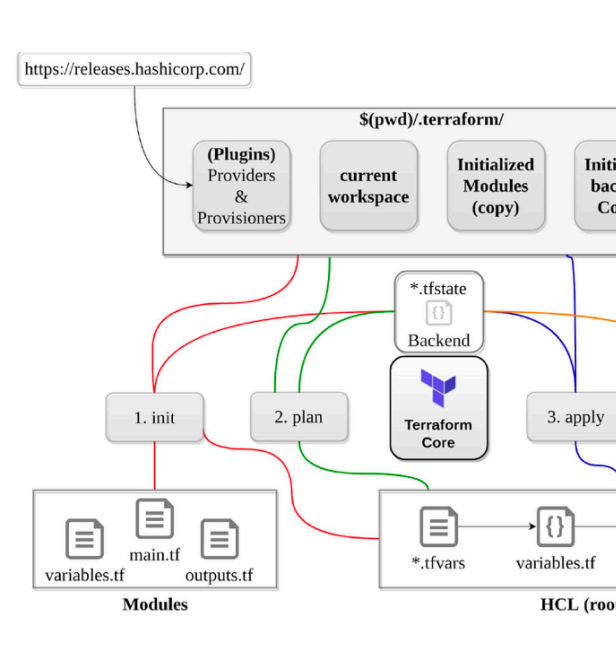
Terraform Provisioners

- run code locally or remotely on resource creation
  - local exec executes code on the machine running terraform
  - remote exec
    - runs on the provisioned resource
    - supports `ssh` and `winrm`
  - requires inline list of commands
- should be used as a last resort
- are defined within the resource block.
- support types – Create and Destroy
  - if creation time fails, resource is tainted if provisioning failed, by default. (next apply it will be re-created)
  - behavior can be overridden by setting the `on_failure` to `continue`, which means ignore and continue
  - for destroy, if it fails – resources are not removed

Terraform Workspaces

- helps manage multiple distinct sets of infrastructure resources or environments with the same code.
- just need to create needed workspace and use them, instead of creating a directory for each environment to manage
- state files for each workspace are stored in the directory `terraform.tfstate.d`
- `terraform workspace new dev` creates a new workspace and switches to it as well
- `terraform workspace select dev` helps select workspace
- `terraform workspace list` lists the workspaces and shows the current active one with `*`
- does not provide strong separation as it uses the same backend

Terraform Workflow



`init`

- initializes a working directory containing Terraform configuration files.
- performs
  - backend initialization , storage for terraform state file.
  - modules installation, downloaded from terraform registry to local path
  - provider(s) plugins installation, the plugins are downloaded in the sub-directory of the present working directory at the path of `.terraform/plugins`
- supports `-upgrade` to update all previously installed plugins to the newest version that complies with the configuration's version constraints
- is safe to run multiple times, to bring the working directory up to date with changes in the configuration
- does not delete the existing configuration or state

`validate`

- validates syntactically for format and correctness.
- is used to validate/check the syntax of the Terraform files.
- verifies whether a configuration is syntactically valid and internally consistent, regardless of any provided variables or existing state.
- A syntax check is done on all the terraform files in the directory, and will display an error if any of the files doesn't validate.

`plan`

- create a execution plan
- traverses each vertex and requests each provider using parallelism
- calculates the difference between the last-known state and the current state and presents this difference as the output of the terraform plan operation to user in their terminal

- does not modify the infrastructure or state.
- allows a user to see which actions Terraform will perform prior to making any changes to reach the desired state
- will scan all `*.tf` files in the directory and create the plan
- will perform refresh for each resource and might hit rate limiting issues as it calls provider APIs
- all resources refresh can be disabled or avoided using
  - `-refresh=false` or
  - `target=xxxx` or
  - break resources into different directories.
- supports `-out` to save the plan

`apply`

- apply changes to reach the desired state.
- scans the current directory for the configuration and applies the changes appropriately.
- can be provided with a explicit plan, saved as out from `terraform plan`
- If no explicit plan file is given on the command line, `terraform apply` will create a new plan automatically and prompt for approval to apply it
- will modify the infrastructure and the state.
- if a resource successfully creates but fails during provisioning,
  - Terraform will error and mark the resource as "tainted".
  - A resource that is tainted has been physically created, but can't be considered safe to use since provisioning failed.
  - Terraform also does not automatically roll back and destroy the resource during the apply when the failure happens, because that would go against the execution plan: the execution plan would've said a resource will be created, but does not say it will ever be deleted.

- does not import any resource.
- supports `-auto-approve` to apply the changes without asking for a confirmation
- supports `-target` to apply a specific module

`refresh`

- used to reconcile the state Terraform knows about (via its state file) with the real-world infrastructure

- does not modify infrastructure, but does modify the state file

`destroy`

- destroy the infrastructure and all resources
- modifies both state and infrastructure
- `terraform destroy -target` can be used to destroy targeted resources
- `terraform plan -destroy` allows creation of destroy plan

`import`

- helps import already-existing external resources, not managed by Terraform, into Terraform state and allow it to manage those resources
- Terraform is not able to auto-generate configurations for those imported modules, for now, and requires you to first write the resource definition in Terraform and then import this resource

`taint`

- marks a Terraform-managed resource as tainted, forcing it to be destroyed and recreated on the next apply.
- will not modify infrastructure, but does modify the state file in order to mark a resource as tainted. Infrastructure and state are changed in next apply.

- can be used to taint a resource within a module

`fmt`

- format to lint the code into a standard format

`console`

- command provides an interactive console for evaluating expressions.

Terraform Modules

- enables code reuse

- supports versioning to maintain compatibility
- stores code remotely
- enables easier testing

- enables encapsulation with all the separate resources under one configuration block
- modules can be nested inside other modules, allowing you to quickly spin up whole separate environments.
- can be referred using `source` attribute
- supports Local and Remote modules
  - Local modules are stored alongside the Terraform configuration (in a separate directory, outside of each environment but in the same repository) with `source` path `./` or `../`
  - Remote modules are stored externally in a separate repository, and supports versioning

- supports following backends
  - Local paths
  - Terraform Registry
  - GitHub

  - Bitbucket
  - Generic Git, Mercurial repositories
  - HTTP URLs

  - S3 buckets
  - GCS buckets

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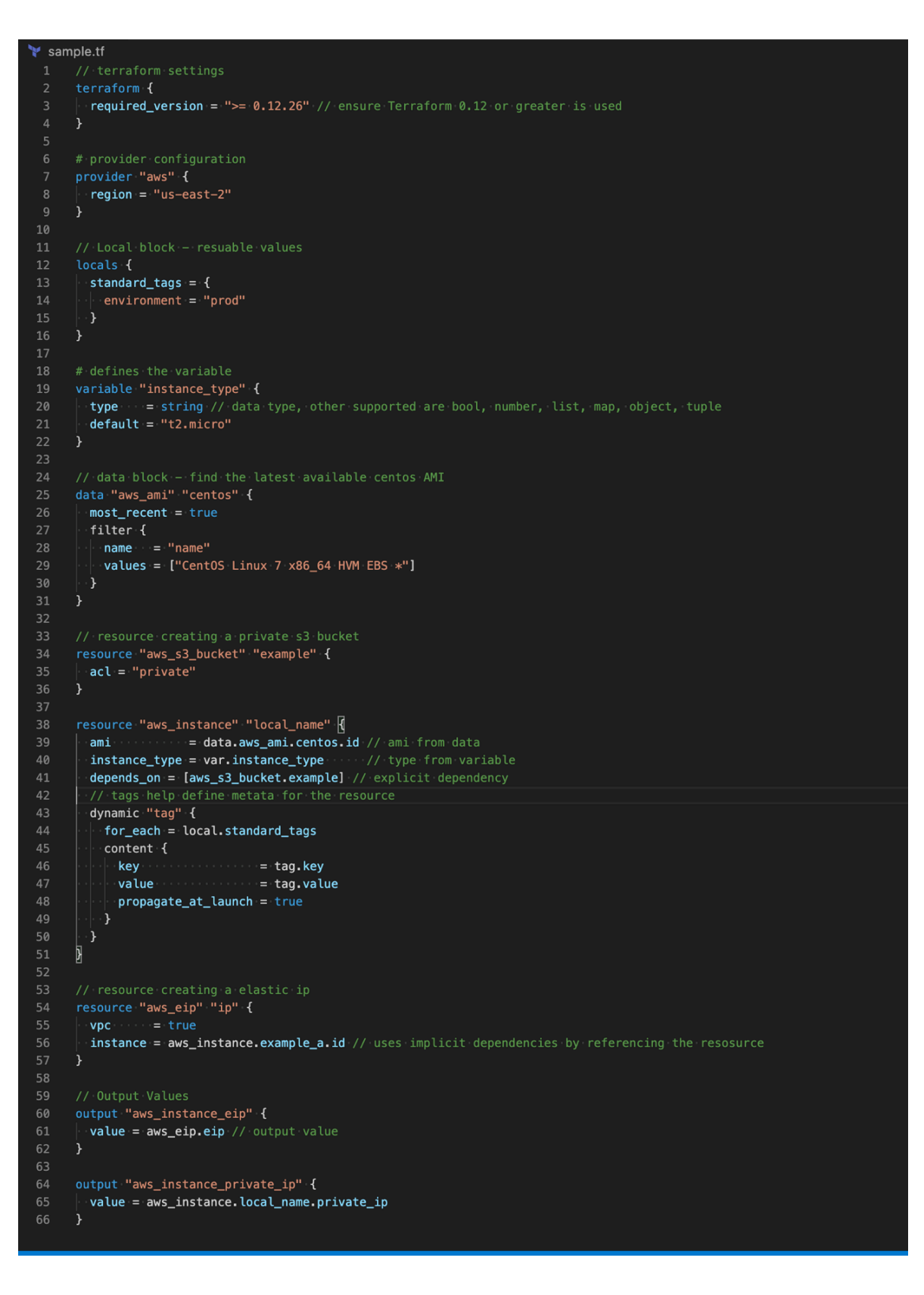
Categories

Select Category

- Module requirements
  - must be on GitHub and must be a public repo, if using public registry.
  - must be named `terraform-<PROVIDER>-<NAME>`, where `<NAME>` reflects the type of infrastructure the module manages and `<PROVIDER>` is the main provider where it creates that infrastructure. *for e.g. terraform-google-vault or terraform-aws-ec2-instance.*
  - must maintain `x.y.z` tags for releases to identify module versions. Release tag names must be a semantic version, which can optionally be prefixed with a v *for example, v1.0.4 and 0.9.2.* Tags that don't look like version numbers are ignored.
  - must maintain a Standard module structure, which allows the registry to inspect the module and generate documentation, track resource usage, parse submodules and examples, and more.

Terraform Read and write configuration



```
// terraform settings
terraform {
  required_version = ">= 0.12.26" // ensure Terraform 0.12 or greater is used
}

# provider configuration
provider "aws" {
  region = "us-east-2"
}

// Local block - resuable values
locals {
  standard_tags = {
    environment = "prod"
  }
}

# defines the variable
variable "instance_type" {
  type    = string // data type, other supported are bool, number, list, map, object, tuple
  default = "t2.micro"
}

// data block - find the latest available centos AMI
data "aws_ami" "centos" {
  most_recent = true
  filter {
    name   = "name"
    values = ["CentOS Linux 7 x86_64 HVM EBS *"]
  }
}

// resource creating a private s3 bucket
resource "aws_s3_bucket" "example" {
  acl = "private"
}

resource "aws_instance" "local_name" {
  ami           = data.aws_ami.centos.id // ami from data
  instance_type = var.instance_type      // type from variable
  depends_on = [aws_s3_bucket.example] // explicit dependency
  // tags help define metata for the resource
  dynamic "tag" {
    for_each = local.standard_tags
    content {
      key                 = tag.key
      value               = tag.value
      propagate_at_launch = true
    }
  }
}

// resource creating a elastic ip
resource "aws_eip" "ip" {
  vpc      = true
  instance = aws_instance.example_a.id // uses implicit dependencies by referencing the resosurce
}

// Output Values
output "aws_instance_eip" {
  value = aws_eip.eip // output value
}

output "aws_instance_private_ip" {
  value = aws_instance.local_name.private_ip
}
```

- **Resources**
  - `resource` are the most important element in the Terraform language that describes one or more infrastructure objects, such as compute instances etc
  - resource type and local name together serve as an identifier for a given resource and must be unique within a module *for e.g.* `aws_instance.local_name`
- **Data Sources**
  - `data` allow data to be fetched or computed for use elsewhere in Terraform configuration
  - allows a Terraform configuration to make use of information defined outside of Terraform, or defined by another separate Terraform configuration
- **Variables**
  - `variable` serve as parameters for a Terraform module and act like function arguments
  - allows aspects of the module to be customized without altering the module's own source code, and allowing modules to be shared between different configurations
  - can be defined through multiple ways
    - command line for e.g.`-var="image_id=ami-abc123"`
    - variable definition files `.tfvars` or `.tfvars.json`. By default, terraform automatically loads
      - Files named exactly `terraform.tfvars` or `terraform.tfvars.json`.
      - Any files with names ending in `.auto.tfvars` or `.auto.tfvars.json`
      - file can also be passed with `-var-file`
    - environment variables can be used to set variables using the format `TF_VAR_name`
  - Environment variables
    - `terraform.tfvars` file, if present.
    - `terraform.tfvars.json` file, if present.
    - Any `*.auto.tfvars` or `*.auto.tfvars.json` files, processed in lexical order of their filenames.
    - Any `-var` and `-var-file` options on the command line, in the order they are provided.Terraform loads variables in the following order, with later sources taking precedence over earlier ones:
- **Local Values**
  - `locals` assigns a name to an expression, allowing it to be used multiple times within a module without repeating it.
  - are like a function's temporary local variables.
  - helps to avoid repeating the same values or expressions multiple times in a configuration.
- **Output**
  - are like function return values.
  - output can be marked as containing sensitive material using the optional `sensitive` argument, which prevents Terraform from showing its value in the list of outputs. However, they are still stored in the state as plain text.
  - In a parent module, outputs of child modules are available in expressions as `module.<MODULE NAME>.<OUTPUT NAME>`.
- **Named Values**
  - is an expression that references the associated value for e.g. `aws_instance.local_name`, `data.aws_ami.centos`, `var.instance_type` etc.
  - support Local named values for e.g `count.index`
- **Dependencies**
  - identifies implicit dependencies as Terraform automatically infers when one resource depends on another by studying the resource attributes used in interpolation expressions for e.g `aws_eip` on resource `aws_instance`
  - explicit dependencies can be defined using `depends_on` where dependencies between resources that are *not* visible to Terraform
- **Data Types**
  - supports primitive data types of
    - `string`, `number` and `bool`
    - Terraform language will automatically convert `number` and `bool` values to `string` values when needed
  - supports complex data types of
    - `list` – a sequence of values identified by consecutive whole numbers starting with zero.
    - `map` – a collection of values where each is identified by a string label.
    - `set` – a collection of unique values that do not have any secondary identifiers or ordering.
  - supports structural data types of
    - `object` – a collection of named attributes that each have their own type
    - `tuple` – a sequence of elements identified by consecutive whole numbers starting with zero, where each element has its own type.
- **Built-in Functions**
  - includes a number of built-in functions that can be called from within expressions to transform and combine values for e.g. `min`, `max`, `file`, `concat`, `element`, `index`, `lookup` etc.
  - does not support user-defined functions
- **Dynamic Blocks**
  - acts much like a `for` expression, but produces nested blocks instead of a complex typed value. It iterates over a given complex value, and generates a nested block for each element of that complex value.
- **Terraform Comments**
  - supports three different syntaxes for comments:
    - #
    - //
    - /* and */

Terraform Backends

- determines how state is loaded and how an operation such as `apply` is executed
- are responsible for storing state and providing an API for optional state locking
- needs to be initialized
- if switching the backed for the first time setup, Terraform provides a migration option
- helps
  - collaboration and working as a team, with the state maintained remotely and state locking
  - can provide enhanced security for sensitive data
  - support remote operations
- supports local vs remote backends
  - local (default) backend stores state in a local JSON file on disk
  - remote backends stores state remotely like S3, OSS, GCS, Consul and support features like remote operation, state locking, encryption, versioning etc.
- supports partial configuration with remaining configuration arguments provided as part of the initialization process
- Backend configuration doesn't support interpolations.
- GitHub is not the supported backend type in Terraform.

Terraform State Management

- state helps keep track of the infrastructure Terraform manages
- stored locally in the `terraform.tfstate`
- recommended not to edit the state manually
- Use `terraform state` command
  - `mv` – to move/rename modules
  - `rm` – to safely remove resource from the state. (destroy/retain like)
  - `pull` – to observe current remote state
  - `list` & `show` – to write/debug modules

State Locking

- happens for all operations that could write state, if supported by backend
- prevents others from acquiring the lock & potentially corrupting the state
- backends which support state locking are
  - azurerm
  - Hashicorp consul
  - Tencent Cloud Object Storage (COS)
  - etcdv3
  - Google Cloud Storage GCS
  - HTTP endpoints
  - Kubernetes Secret with locking done using a Lease resource
  - AliCloud Object Storage OSS with locking via TableStore
  - PostgreSQL
  - AWS S3 with locking via DynamoDB
  - Terraform Enterprise
- Backends which do not support state locking are
  - artifactory
  - etcd
- can be disabled for most commands with the `-lock` flag
- use `force-unlock` command to manually unlock the state if unlocking failed

State Security

- can contain sensitive data, depending on the resources in use *for e.g passwords and keys*
- using local state, data is stored in plain-text JSON files
- using remote state, state is held in memory when used by Terraform. It may be encrypted at rest, if supported by backend *for e.g. S3, OSS*

Terraform Logging

- debugging can be controlled using `TF_LOG`, which can be configured for different levels TRACE, DEBUG, INFO, WARN or ERROR, with TRACE being the more verbose.
- logs path can be controlled `TF_LOG_PATH`. `TF_LOG` needs to be specified.

Terraform Cloud and Terraform Enterprise

- Terraform Cloud provides Cloud Infrastructure Automation as a Service. It is offered as a multi-tenant SaaS platform and is designed to suit the needs of smaller teams and organizations. Its smaller plans default to one run at a time, which prevents users from executing multiple runs concurrently.
- Terraform Enterprise is a private install for organizations who prefer to self-manage. It is designed to suit the needs of organizations with specific requirements for security, compliance and custom operations.
- Terraform Cloud provides features
  - **Remote Terraform Execution** – supports Remote Operations for Remote Terraform execution which helps provide consistency and visibility for critical provisioning operations.
  - **Workspaces** – organizes infrastructure with workspaces instead of directories. Each workspace contains everything necessary to manage a given collection of infrastructure, and Terraform uses that content whenever it executes in the context of that workspace.
  - **Remote State Management** – acts as a remote backend for the Terraform state. State storage is tied to workspaces, which helps keep state associated with the configuration that created it.
  - **Version Control Integration –** is designed to work directly with the version control system (VCS) provider.
  - **Private Module Registry** – provides a private and central library of versioned & validated modules to be used within the organization
  - **Team based Permission System** – can define groups of users that match the organization's real-world teams and assign them only the permissions they need
  - **Sentinel Policies** – embeds the Sentinel policy-as-code framework, which lets you define and enforce granular policies for how the organization provisions infrastructure. Helps eliminate provisioned resources that don't follow security, compliance, or operational policies.
  - **Cost Estimation** – can display an estimate of its total cost, as well as any change in cost caused by the proposed updates
  - Security – encrypts state at rest and protects it with TLS in transit.
- Terraform Enterprise features
  - includes all the Terraform Cloud features with
  - **Audit** – supports detailed audit logging and tracks the identity of the user requesting state and maintains a history of state changes.
  - **SSO/SAML** – SAML for SSO provides the ability to govern user access to your applications.
- Terraform Enterprise currently supports running under the following operating systems for a Clustered deployment:
  - Ubuntu 16.04.3 – 16.04.5 / 18.04
  - Red Hat Enterprise Linux 7.4 through 7.7
  - CentOS 7.4 – 7.7
  - Amazon Linux
  - Oracle Linux
  - Clusters currently don't support other Linux variants.
- Terraform Cloud currently supports following VCS Provider
  - GitHub.com
  - GitHub.com (OAuth)
  - GitHub Enterprise
  - GitLab.com
  - GitLab EE and CE
  - Bitbucket Cloud
  - Bitbucket Server
  - Azure DevOps Server
  - Azure DevOps Services
- A Terraform Enterprise install that is provisioned on a network that does not have Internet access is generally known as an **air-gapped install**. These types of installs require you to pull updates, providers, etc. from external sources vs. being able to download them directly.

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8 thoughts on "Terraform Cheat Sheet"

Surender Aireddy
JANUARY 25, 2021 AT 5:44 AM

Thank you for putting together excellent notes for Terraform!

jayendrapatil
JANUARY 25, 2021 AT 10:58 AM

Glads its helping Surender …

vamsi
FEBRUARY 23, 2021 AT 4:40 PM

sir any suggestions on terraform associate exam i had scheduled the exam and about to write the exam in few days (i am a beginner)

vamsi
FEBRUARY 23, 2021 AT 4:41 PM

sir any suggestions on terraform associate exam i had scheduled the exam and about to write the exam in few days (i am a beginner)

jayendrapatil
FEBRUARY 26, 2021 AT 11:03 AM

hi vamsi, i have documented the same at https://jayendrapatil.com/hashicorp-certified-terraform-associate-learning-path/.

Raman
APRIL 16, 2021 AT 1:57 AM

Thanks for sharing detailed summary details for Terraform

Navneet Singh
MAY 16, 2021 AT 2:02 PM

Thanks A lot! It helped me alot in Interview.

jayendrapatil
JUNE 6, 2021 AT 6:07 AM

Great it helped you Navneet …

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